

SPX test-set forecast accuracy (\* significant at 5%)

| Model     | Test log-RMSE | Test QLIKE | DM p vs HAR-RV (SE) |
|-----------|---------------|------------|---------------------|
| GARCH     | 0.884         | 0.454      | 0.000*              |
| HAR-RV    | 0.835         | 0.445      | -                   |
| TCNN-LSTM | 0.823         | 0.424      | 0.013*              |